

# T3 UV one-loop RGE term comparison

## Notation key

| symbol                    | interaction / definition                                           |
|---------------------------|--------------------------------------------------------------------|
| $g_1$                     | $D_\mu \supset ig_1 Y B_\mu$                                       |
| $g_2$                     | $D_\mu \supset ig_2 T^A W_\mu^A$                                   |
| $g_3$                     | $D_\mu \supset ig_3 t^A G_\mu^A$                                   |
| $Y_u$                     | $-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$                           |
| $Y_d$                     | $-\bar{Q} Y_d H d + \text{h.c.}$                                   |
| $Y_e$                     | $-\bar{L} Y_e H e + \text{h.c.}$                                   |
| $y_1$                     | $LF S_1 + \text{h.c.}$                                             |
| $y_2$                     | $LF S_2 + \text{h.c.}$                                             |
| $M_F$                     | $-\bar{F} M_F F$                                                   |
| $T$                       | $\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$             | $\text{Tr}(y_1^\dagger y_1)$                                       |
| $T_\nu^{(2)}$             | $\text{Tr}(y_2^\dagger y_2)$                                       |
| $T_\nu$                   | $T_\nu^{(1)} + T_\nu^{(2)}$                                        |
| $m_1^2$                   | $m_1^2 S_1^\dagger S_1$                                            |
| $m_2^2$                   | $m_2^2 S_2^\dagger S_2$                                            |
| $\lambda_1$               | $\frac{1}{2} \lambda_1 (H^\dagger H)^2$                            |
| $\lambda_{S_1}^{(1)}$     | $\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$              |
| $\lambda_{S_2}^{(1)}$     | $\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$              |
| $\lambda_{HS_1}^{(1)}$    | $\lambda_{HS_1}^{(1)} (H^\dagger H) (S_1^\dagger S_1)$             |
| $\lambda_{HS_2}^{(1)}$    | $\lambda_{HS_2}^{(1)} (H^\dagger H) (S_2^\dagger S_2)$             |
| $\lambda_{HS_1}^{(A)}$    | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H) (S_1^\dagger T^A S_1)$     |
| $\lambda_{HS_2}^{(A)}$    | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H) (S_2^\dagger T^A S_2)$     |
| $\lambda_5$               | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$                      |
| $\lambda_{12}^{(1)}$      | $\lambda_{12}^{(1)} (S_1^\dagger S_1) (S_2^\dagger S_2)$           |
| $\lambda_{12}^{(A)}$      | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1) (S_2^\dagger T^A S_2)$   |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2) (S_2^\dagger S_1)$      |
| $\lambda_{S_1}^{(A)}$     | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1) (S_1^\dagger T^A S_1)$  |
| $\lambda_{S_2}^{(A)}$     | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2) (S_2^\dagger T^A S_2)$  |
| $\alpha$                  | $Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$   |
| $C_2(d)$                  | $(d^2 - 1)/4$                                                      |
| $T(d)$                    | $d(d^2 - 1)/12$                                                    |
| $C_{LLS_1 S_2}^{ij;AB}$   | $L_i L_j S_1^A S_2^B$                                              |
| $C_{ijab}$                | $L_i L_j \phi_a \phi_b$                                            |
| $C_5^{ij}$                | $(L_i^T C L_j) H H$                                                |

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use  $16\pi^2\,\mu\,dX/d\mu=\beta_X^{(1)}$ . Gauge-coupling results use the same RGBeta  $g^2\rightarrow g$  conversion as in the previous report.

$\beta_{g_1}^{(1)}$

Quantity:  $g_1$

$16\pi^2\,\mu\frac{dg_1}{d\mu}=\beta_{g_1}^{(1)}$

| Column Representative term structure |                     |
|--------------------------------------|---------------------|
| T1                                   | $\frac{25g_1^3}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                  |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|---------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $\frac{25g_1^3}{2}$ |

$\beta_{g_2}^{(1)}$   
Quantity:  $g_2$

$$16\pi^2\,\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-\frac{17g_2^3}{6}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                   |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|----------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-\frac{17g_2^3}{6}$ |

$\beta_{g_3}^{(1)}$   
Quantity:  $g_3$

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-7g_3^3$                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1        |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-7g_3^3$ |

$\beta_{Y_u}^{(1)}$

Quantity:  $Y_u$

$$16\pi^2\,\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                             |
|--------|-----------------------------------------------------------|
| T1     | $-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$ |
| T2     | $+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$   |
| T3     | $+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                       | T2                                                      | T3                                                        |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|----------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$ | $+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$ |

Quantity:  $Y_u$

Terms 2 of 3

| Column | Representative term structure                                                   |
|--------|---------------------------------------------------------------------------------|
| T1     | $-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$ |
| T2     | $-\frac{9}{4}g_2^2\ (Y_u)_{ij}$                                                 |
| T3     | $-8g_3^2\ (Y_u)_{ij}$                                                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                                              | T2                              | T3                    |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|---------------------------------------------------------------------------------|---------------------------------|-----------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$ | $-\frac{9}{4}g_2^2\ (Y_u)_{ij}$ | $-8g_3^2\ (Y_u)_{ij}$ |

Quantity:  $Y_u$

Terms 3 of 3

| Column | Representative term structure   |
|--------|---------------------------------|
| T1     | $-\frac{17}{12}g_1^2(Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                              |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|---------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-\frac{17}{12}g_1^2(Y_u)_{ij}$ |

$\beta_{Y_d}^{(1)}$

**Quantity:**  $Y_d$

$$16\pi^2\,\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$$

Terms 1 of 3

| Column | Representative term structure                           |
|--------|---------------------------------------------------------|
| T1     | $\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$     |
| T2     | $+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$ |
| T3     | $+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                  | T2                                                      | T3                                                      |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$ | $+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$ | $+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$ |



Quantity:  $Y_d$

Terms 2 of 3

| Column | Representative term structure                                                 |
|--------|-------------------------------------------------------------------------------|
| T1     | $+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$ |
| T2     | $-\frac{9}{4}g_2^2\ (Y_d)_{ij}$                                               |
| T3     | $-8g_3^2\ (Y_d)_{ij}$                                                         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                                            | T2                              | T3                    |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------------------------------------------------------|---------------------------------|-----------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$ | $-\frac{9}{4}g_2^2\ (Y_d)_{ij}$ | $-8g_3^2\ (Y_d)_{ij}$ |

Quantity:  $Y_d$

Terms 3 of 3

| Column | Representative term structure  |
|--------|--------------------------------|
| T1     | $-\frac{5}{12}g_1^2(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                             |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-\frac{5}{12}g_1^2(Y_d)_{ij}$ |

$\beta_{Y_e}^{(1)}$   
Quantity:  $Y_e$

$16\pi^2\mu\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 2

| Column | Representative term structure                           |
|--------|---------------------------------------------------------|
| T1     | $-3\mathrm{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_e)_{ij}$ |
| T2     | $+3\mathrm{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_e)_{ij}$ |
| T3     | $+\mathrm{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_e)_{ij}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                     | T2                                                      | T3                                                     |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $3\mathrm{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_e)_{ij}$ | $+3\mathrm{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_e)_{ij}$ | $+\mathrm{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_e)_{ij}$ |

Quantity:  $Y_e$

Terms 2 of 2

| Column | Representative term structure                                                                                   |
|--------|-----------------------------------------------------------------------------------------------------------------|
| T1     | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T2     | $-\frac{9}{4}g_2^2\ (Y_e)_{ij}$                                                                                 |
| T3     | $-\frac{15}{4}g_1^2\ (Y_e)_{ij}$                                                                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                                                                              | T2                              | T3                               |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ | $-\frac{9}{4}g_2^2\ (Y_e)_{ij}$ | $-\frac{15}{4}g_1^2\ (Y_e)_{ij}$ |

$\beta_{y_1}^{(1)}$

Quantity:  $y_1$

$$16\pi^2\,\mu\frac{dy_1}{d\mu}=\beta_{y_1}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                                                                         |
|--------|-----------------------------------------------------------------------------------------------------------------------|
| T1     | $T_{\nu}^{(1)}(y_1)_{ij}$                                                                                             |
| T2     | $+\frac{1}{2}((y_2\ (y_2)^{\dagger}\ y_1)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_1)_{ij}+3\,(y_1\ (y_1)^{\dagger}\ y_1)_{ij})$ |
| T3     | $-\frac{9}{4}g_2^2\,(y_1)_{ij}$                                                                                       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                        | T2                                                                                                                    | T3                              |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|---------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $T_{\nu}^{(1)}(y_1)_{ij}$ | $+\frac{1}{2}((y_2\ (y_2)^{\dagger}\ y_1)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_1)_{ij}+3\,(y_1\ (y_1)^{\dagger}\ y_1)_{ij})$ | $-\frac{9}{4}g_2^2\,(y_1)_{ij}$ |

Quantity:  $y_1$

Terms 2 of 2

| Column | Representative term structure  |
|--------|--------------------------------|
| T1     | $-\frac{15}{4}g_1^2(y_1)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                             |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-\frac{15}{4}g_1^2(y_1)_{ij}$ |

$\beta_{y_2}^{(1)}$   
Quantity:  $y_2$

$$16\pi^2\,{}_{\mu}\frac{dy_2}{d\mu}=\beta_{y_2}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                                                                         |
|--------|-----------------------------------------------------------------------------------------------------------------------|
| T1     | $T_{\nu}^{(2)}(y_2)_{ij}$                                                                                             |
| T2     | $+\frac{1}{2}((y_1\ (y_1)^{\dagger}\ y_2)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_2)_{ij}+3\ (y_2\ (y_2)^{\dagger}\ y_2)_{ij})$ |
| T3     | $-\frac{9}{4}g_2^2\ (y_2)_{ij}$                                                                                       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                        | T2                                                                                                                    | T3                              |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|---------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $T_{\nu}^{(2)}(y_2)_{ij}$ | $+\frac{1}{2}((y_1\ (y_1)^{\dagger}\ y_2)_{ij}+(Y_e\ (Y_e)^{\dagger}\ y_2)_{ij}+3\ (y_2\ (y_2)^{\dagger}\ y_2)_{ij})$ | $-\frac{9}{4}g_2^2\ (y_2)_{ij}$ |

Quantity:  $y_2$

Terms 2 of 2

| Column | Representative term structure  |
|--------|--------------------------------|
| T1     | $-\frac{15}{4}g_1^2(y_2)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                             |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-\frac{15}{4}g_1^2(y_2)_{ij}$ |



$\beta_{M_F}^{(1)}$

Quantity:  $M_F$

$16\pi^2\mu\frac{dM_F}{d\mu}=\beta_{M_F}^{(1)}$

| Column | Representative term structure                 |
|--------|-----------------------------------------------|
| T1     | $\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$     |
| T2     | $+ \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$ |
| T3     | $-6g_1^2\ (M_F)_{ij}$                         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                        | T2                                            | T3                    |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------------------|-----------------------------------------------|-----------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$ | $+ \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$ | $-6g_1^2\ (M_F)_{ij}$ |

$\beta_{m_1^2}^{(1)}$

Quantity:  $m_1^2$

$16\pi^2\mu\frac{dm_1^2}{d\mu}=\beta_{m_1^2}^{(1)}$

Terms 1 of 2

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$                    |
| T2     | $-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$ |
| T3     | $-\frac{9g_2^2m_1^2}{2}$                                      |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                         | T2                                                            | T3                       |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------------------------------------|---------------------------------------------------------------|--------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$ | $-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$ | $-\frac{9g_2^2m_1^2}{2}$ |

Quantity:  $m_1^2$

Terms 2 of 2

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-\frac{3g_1^2m_1^2}{2}$      |
| T2     | $+4\lambda_{12}^{(1)}m_2^2$   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                       | T2                          |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------------------|-----------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-\frac{3g_1^2m_1^2}{2}$ | $+4\lambda_{12}^{(1)}m_2^2$ |

$\beta_{m_2^2}^{(1)}$

Quantity:  $m_2^2$

$16\pi^2\mu\frac{dm_2^2}{d\mu}=\beta_{m_2^2}^{(1)}$

Terms 1 of 2

| Column | Representative term structure                                   |
|--------|-----------------------------------------------------------------|
| T1     | $m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$                      |
| T2     | $-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$ |
| T3     | $-\frac{9g_2^2m_2^2}{2}$                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                         | T2                                                              | T3                       |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------------------------------------|-----------------------------------------------------------------|--------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$ | $-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$ | $-\frac{9g_2^2m_2^2}{2}$ |

Quantity:  $m_2^2$

Terms 2 of 2

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-\frac{27g_1^2m_2^2}{2}$     |
| T2     | $+4\lambda_{12}^{(1)}m_1^2$   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                        | T2                          |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|---------------------------|-----------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-\frac{27g_1^2m_2^2}{2}$ | $+4\lambda_{12}^{(1)}m_1^2$ |

$\beta_{\lambda_1}^{(1)}$   
Quantity:  $\lambda_1$

$$16\pi^2\,\mu\,\frac{d\lambda_1}{d\mu}=\beta_{\lambda_1}^{(1)}$$

Terms 1 of 5

| Column | Representative term structure                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |
| T2     | $-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$                                                                          |
| T3     | $-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$                                                                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                                                                                                        | T2                                                               | T3                                                              |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$ | $-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ |

Quantity:  $\lambda_1$

Terms 2 of 5

| Column | Representative term structure                                      |
|--------|--------------------------------------------------------------------|
| T1     | $-12\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\,Y_u\,(Y_u)^\dagger\Big)$ |
| T2     | $+4\lambda_5\,(\lambda_5)^*$                                       |
| T3     | $+ \frac{9g_2^4}{4}$                                               |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                                 | T2                           | T3                   |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------------------------------------------------------------|------------------------------|----------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-12\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\,Y_u\,(Y_u)^\dagger\Big)$ | $+4\lambda_5\,(\lambda_5)^*$ | $+ \frac{9g_2^4}{4}$ |

Quantity:  $\lambda_1$

Terms 3 of 5

| Column | Representative term structure                    |
|--------|--------------------------------------------------|
| T1     | $+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$ |
| T2     | $-9g_2^2\lambda_1$                               |
| T3     | $+\frac{3g_1^4}{4}$                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                               | T2                 | T3                  |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------------------------------------------|--------------------|---------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$ | $-9g_2^2\lambda_1$ | $+\frac{3g_1^4}{4}$ |



Quantity:  $\lambda_1$

Terms 4 of 5

| Column | Representative term structure      |
|--------|------------------------------------|
| T1     | $+12\lambda_1^2$                   |
| T2     | $+4\lambda_{HS_1}^{(1)2}$          |
| T3     | $+\frac{\lambda_{HS_1}^{(A)2}}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1               | T2                        | T3                                 |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|------------------|---------------------------|------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+12\lambda_1^2$ | $+4\lambda_{HS_1}^{(1)2}$ | $+\frac{\lambda_{HS_1}^{(A)2}}{4}$ |

Terms 5 of 5

| Column | Representative term structure       |
|--------|-------------------------------------|
| T1     | $+4\lambda_{HS_2}^{(1)^2}$          |
| T2     | $+\frac{\lambda_{HS_2}^{(A)^2}}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                         | T2                                  |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|----------------------------|-------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+4\lambda_{HS_2}^{(1)^2}$ | $+\frac{\lambda_{HS_2}^{(A)^2}}{4}$ |

$$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$$

Quantity:  $\lambda_{S_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_1}^{(1)}}{d\mu}=\beta_{\lambda_{S_1}^{(1)}}^{(1)}$$

### Terms 1 of 4

| Column | Representative term structure                                         |
|--------|-----------------------------------------------------------------------|
| T1     | $4\lambda_{S_1}^{(1)}T_{\nu}^{(1)}$                                   |
| T2     | $-4\,\mathrm{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_1\,(y_1)^{\dagger}\Big)$ |
| T3     | $+\frac{9g_2^4}{4}$                                                   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                  | T2                                                                    | T3                  |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------------|-----------------------------------------------------------------------|---------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $4\lambda_{S_1}^{(1)}T_{\nu}^{(1)}$ | $-4\,\mathrm{Tr}\Big(y_1\,(y_1)^{\dagger}\,y_1\,(y_1)^{\dagger}\Big)$ | $+\frac{9g_2^4}{4}$ |

Quantity:  $\lambda_{S_1}^{(1)}$

Terms 2 of 4

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$ |
| T2     | $-9g_2^2\lambda_{S_1}^{(1)}$                               |
| T3     | $+\frac{3g_1^4}{4}$                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                         | T2                           | T3                  |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|------------------------------------------------------------|------------------------------|---------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$ | $-9g_2^2\lambda_{S_1}^{(1)}$ | $+\frac{3g_1^4}{4}$ |

Quantity:  $\lambda_{S_1}^{(1)}$

Terms 3 of 4

| Column | Representative term structure    |
|--------|----------------------------------|
| T1     | $+4\lambda_{12}^{(1)2}$          |
| T2     | $+\frac{\lambda_{12}^{(A)2}}{4}$ |
| T3     | $+4\lambda_{HS_1}^{(1)2}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                      | T2                               | T3                        |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------|----------------------------------|---------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+4\lambda_{12}^{(1)2}$ | $+\frac{\lambda_{12}^{(A)2}}{4}$ | $+4\lambda_{HS_1}^{(1)2}$ |

Quantity:  $\lambda_{S_1}^{(1)}$

Terms 4 of 4

| Column | Representative term structure       |
|--------|-------------------------------------|
| T1     | $+\frac{\lambda_{HS_1}^{(A)^2}}{4}$ |
| T2     | $+12\lambda_{S_1}^{(1)^2}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                  | T2                         |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------------|----------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+\frac{\lambda_{HS_1}^{(A)^2}}{4}$ | $+12\lambda_{S_1}^{(1)^2}$ |

$\beta_{\lambda_{S_2}^{(1)}}^{(1)}$

Quantity:  $\lambda_{S_2}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_2}^{(1)}}{d\mu}=\beta_{\lambda_{S_2}^{(1)}}^{(1)}$

Terms 1 of 4

| Column | Representative term structure                                                                |
|--------|----------------------------------------------------------------------------------------------|
| T1     | $4\lambda_{S_2}^{(1)}T_\nu^{(2)}$                                                            |
| T2     | $-4\operatorname{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ |
| T3     | $+\frac{9g_2^4}{4}$                                                                          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                | T2                                                                                           | T3                  |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------------------|----------------------------------------------------------------------------------------------|---------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $4\lambda_{S_2}^{(1)}T_\nu^{(2)}$ | $-4\operatorname{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ | $+\frac{9g_2^4}{4}$ |

Quantity:  $\lambda_{S_2}^{(1)}$

Terms 2 of 4

| Column | Representative term structure                                |
|--------|--------------------------------------------------------------|
| T1     | $+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)}\right)$ |
| T2     | $-9g_2^2\lambda_{S_2}^{(1)}$                                 |
| T3     | $+\frac{243g_1^4}{4}$                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                           | T2                           | T3                    |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------------------------------------------------------|------------------------------|-----------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)}\right)$ | $-9g_2^2\lambda_{S_2}^{(1)}$ | $+\frac{243g_1^4}{4}$ |



Quantity:  $\lambda_{S_2}^{(1)}$

Terms 3 of 4

| Column | Representative term structure    |
|--------|----------------------------------|
| T1     | $+4\lambda_{12}^{(1)2}$          |
| T2     | $+\frac{\lambda_{12}^{(A)2}}{4}$ |
| T3     | $+4\lambda_{HS_2}^{(1)2}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                      | T2                               | T3                        |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------|----------------------------------|---------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+4\lambda_{12}^{(1)2}$ | $+\frac{\lambda_{12}^{(A)2}}{4}$ | $+4\lambda_{HS_2}^{(1)2}$ |

Quantity:  $\lambda_{S_2}^{(1)}$

Terms 4 of 4

| Column | Representative term structure       |
|--------|-------------------------------------|
| T1     | $+\frac{\lambda_{HS_2}^{(A)^2}}{4}$ |
| T2     | $+12\lambda_{S_2}^{(1)^2}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                  | T2                         |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------------|----------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+\frac{\lambda_{HS_2}^{(A)^2}}{4}$ | $+12\lambda_{S_2}^{(1)^2}$ |

$$\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

Quantity:  $\lambda_{HS_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(1)}}{d\mu}=\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

### Terms 1 of 4

| Column | Representative term structure                                                                                                                                                                                        |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{HS_1}^{(1)}\left(2T_\nu^{(1)}+6\,\text{Tr}\!\left(Y_d\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\!\left(Y_e\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\!\left(Y_u\left(Y_u\right)^\dagger\right)\right)$ |
| T2     | $-2\,\text{Tr}\!\left(y_1\left(y_1\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$                                                                                                                                |
| T3     | $+6\lambda_5\left(\lambda_5\right)^*$                                                                                                                                                                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                                                                                                                                                                                   | T2                                                                                    | T3                                    |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $\lambda_{HS_1}^{(1)}\left(2T_\nu^{(1)}+6\,\text{Tr}\!\left(Y_d\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\!\left(Y_e\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\!\left(Y_u\left(Y_u\right)^\dagger\right)\right)$ | $-2\,\text{Tr}\!\left(y_1\left(y_1\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ | $+6\lambda_5\left(\lambda_5\right)^*$ |

Quantity:  $\lambda_{HS_1}^{(1)}$

Terms 2 of 4

| Column | Representative term structure    |
|--------|----------------------------------|
| T1     | $+ \frac{9 g_2^4}{4}$            |
| T2     | $- 9 g_2^2 \lambda_{HS_1}^{(1)}$ |
| T3     | $+ \frac{3 g_1^4}{4}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                    | T2                               | T3                    |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------|----------------------------------|-----------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+ \frac{9 g_2^4}{4}$ | $- 9 g_2^2 \lambda_{HS_1}^{(1)}$ | $+ \frac{3 g_1^4}{4}$ |

Quantity:  $\lambda_{HS_1}^{(1)}$

Terms 3 of 4

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $-3g_1^2\lambda_{HS_1}^{(1)}$              |
| T2     | $+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$ |
| T3     | $+6\lambda_1\lambda_{HS_1}^{(1)}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                            | T2                                         | T3                                |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------|--------------------------------------------|-----------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-3g_1^2\lambda_{HS_1}^{(1)}$ | $+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$ | $+6\lambda_1\lambda_{HS_1}^{(1)}$ |

Quantity:  $\lambda_{HS_1}^{(1)}$

Terms 4 of 4

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $+4\lambda_{HS_1}^{(1)\ 2}$                 |
| T2     | $+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$ |
| T3     | $+\frac{3\lambda_{HS_1}^{(A)\ 2}}{4}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                          | T2                                          | T3                                    |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------------|---------------------------------------------|---------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+4\lambda_{HS_1}^{(1)\ 2}$ | $+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$ | $+\frac{3\lambda_{HS_1}^{(A)\ 2}}{4}$ |

$$\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$$

Quantity:  $\lambda_{HS_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(1)}}{d\mu}=\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$$

### Terms 1 of 4

| Column | Representative term structure                                                                                                                                   |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |
| T2     | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$                                                                                                 |
| T3     | $+6\lambda_5\,(\lambda_5)^*$                                                                                                                                    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                                                                                                                              | T2                                                              | T3                           |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+6\lambda_5\,(\lambda_5)^*$ |

Quantity:  $\lambda_{HS_2}^{(1)}$

Terms 2 of 4

| Column | Representative term structure    |
|--------|----------------------------------|
| T1     | $+ \frac{9 g_2^4}{4}$            |
| T2     | $- 9 g_2^2 \lambda_{HS_2}^{(1)}$ |
| T3     | $+ \frac{27 g_1^4}{4}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                    | T2                               | T3                     |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------|----------------------------------|------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+ \frac{9 g_2^4}{4}$ | $- 9 g_2^2 \lambda_{HS_2}^{(1)}$ | $+ \frac{27 g_1^4}{4}$ |



Quantity:  $\lambda_{HS_2}^{(1)}$

Terms 3 of 4

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $-15g_1^2\lambda_{HS_2}^{(1)}$             |
| T2     | $+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$ |
| T3     | $+6\lambda_1\lambda_{HS_2}^{(1)}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                             | T2                                         | T3                                |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|--------------------------------|--------------------------------------------|-----------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-15g_1^2\lambda_{HS_2}^{(1)}$ | $+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$ | $+6\lambda_1\lambda_{HS_2}^{(1)}$ |

Quantity:  $\lambda_{HS_2}^{(1)}$

Terms 4 of 4

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $+4\lambda_{HS_2}^{(1)\ 2}$                 |
| T2     | $+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3     | $+\frac{3\lambda_{HS_2}^{(A)\ 2}}{4}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                          | T2                                          | T3                                    |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------------|---------------------------------------------|---------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+4\lambda_{HS_2}^{(1)\ 2}$ | $+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$ | $+\frac{3\lambda_{HS_2}^{(A)\ 2}}{4}$ |

$\beta_{\lambda_{12}^{(1)}}^{(1)}$

Quantity:  $\lambda_{12}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{12}^{(1)}}{d\mu}=\beta_{\lambda_{12}^{(1)}}^{(1)}$

Terms 1 of 4

| Column | Representative term structure                                                                |
|--------|----------------------------------------------------------------------------------------------|
| T1     | $\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)})$                                              |
| T2     | $-2\operatorname{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ |
| T3     | $+3\lambda_5\left(\lambda_5\right)^*$                                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                              | T2                                                                                           | T3                                    |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)})$ | $-2\operatorname{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ | $+3\lambda_5\left(\lambda_5\right)^*$ |

Quantity:  $\lambda_{12}^{(1)}$

Terms 2 of 4

| Column | Representative term structure  |
|--------|--------------------------------|
| T1     | $+ \frac{9 g_2^4}{4}$          |
| T2     | $- 9 g_2^2 \lambda_{12}^{(1)}$ |
| T3     | $+ \frac{27 g_1^4}{4}$         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                    | T2                             | T3                     |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------|--------------------------------|------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+ \frac{9 g_2^4}{4}$ | $- 9 g_2^2 \lambda_{12}^{(1)}$ | $+ \frac{27 g_1^4}{4}$ |

Quantity:  $\lambda_{12}^{(1)}$

Terms 3 of 4

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $-15g_1^2\lambda_{12}^{(1)}$              |
| T2     | $+4\lambda_{12}^{(1)2}$                   |
| T3     | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                           | T2                      | T3                                        |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|------------------------------|-------------------------|-------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $-15g_1^2\lambda_{12}^{(1)}$ | $+4\lambda_{12}^{(1)2}$ | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ |

Quantity:  $\lambda_{12}^{(1)}$

Terms 4 of 4

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$    |
| T2     | $+\frac{3\lambda_{12}^{(A)2}}{4}$            |
| T3     | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                        | T2                                | T3                                           |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------------------|-----------------------------------|----------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ | $+\frac{3\lambda_{12}^{(A)2}}{4}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

$\beta_{\lambda_5}^{(1)}$   
Quantity:  $\lambda_5$

$$16\pi^2\,\mu\,\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 3

| Column | Representative term structure                                                                                                                                                                                                                                                                               |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$ |
| T2     | $-9g_2^2\lambda_5$                                                                                                                                                                                                                                                                                          |
| T3     | $-9g_1^2\lambda_5$                                                                                                                                                                                                                                                                                          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                                                                                                                                                                                                                                                                          | T2                 | T3                 |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS1}^{(A)}}{3}+2\lambda_{HS2}^{(A)}\right)$ | $-9g_2^2\lambda_5$ | $-9g_1^2\lambda_5$ |

Quantity:  $\lambda_5$

Terms 2 of 3

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $+2\lambda_{12}^{(1)}\lambda_5$   |
| T2     | $+2\lambda_1\lambda_5$            |
| T3     | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                              | T2                     | T3                                |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|---------------------------------|------------------------|-----------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |



Quantity:  $\lambda_5$

Terms 3 of 3

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $+4\lambda_{HS_2}^{(1)}\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+4\lambda_{HS_2}^{(1)}\lambda_5$ |

$\beta_{\lambda_{12}^{(A)}}^{(1)}$

Quantity:  $\lambda_{12}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{(A)}}{d\mu}=\beta_{\lambda_{12}^{(A)}}^{(1)}$$

Terms 1 of 3

| Column | Representative term structure                                               |
|--------|-----------------------------------------------------------------------------|
| T1     | $\lambda_{12}^{(A)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$                         |
| T2     | $-\frac{8}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$ |
| T3     | $+\frac{4}{3}\lambda_5\ (\lambda_5)^*$                                      |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                  | T2                                                                          | T3                                     |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $\lambda_{12}^{(A)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$ | $-\frac{8}{3}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$ | $+\frac{4}{3}\lambda_5\ (\lambda_5)^*$ |

Quantity:  $\lambda_{12}^{(A)}$

Terms 2 of 3

| Column | Representative term structure                     |
|--------|---------------------------------------------------|
| T1     | $+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$ |
| T2     | $-9g_2^2\lambda_{12}^{(A)}$                       |
| T3     | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                | T2                          | T3                                       |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|---------------------------------------------------|-----------------------------|------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$ | $-9g_2^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |

Quantity:  $\lambda_{12}^{(A)}$

Terms 3 of 3

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$   |
| T2     | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$   |
| T3     | $+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                        | T2                                        | T3                                          |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------------------|-------------------------------------------|---------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | $+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |

$$\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

Quantity:  $\lambda_{HS_1}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(A)}}{d\mu}=\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                                                                                                                                                                                     |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{HS_1}^{(A)}\left(2T_\nu^{(1)}+6\,\mathrm{Tr}\Big(Y_d\,\left(Y_d\right)^\dagger\Big)+2\,\mathrm{Tr}\Big(Y_e\,\left(Y_e\right)^\dagger\Big)+6\,\mathrm{Tr}\Big(Y_u\,\left(Y_u\right)^\dagger\Big)\right)$ |
| T2     | $+\frac{8}{3}\,\mathrm{Tr}\Big(y_1\,\left(y_1\right)^\dagger\,Y_e\,\left(Y_e\right)^\dagger\Big)$                                                                                                                 |
| T3     | $+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$                                                                                                                                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                                                                                                                                                                                | T2                                                                                                | T3                                               |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $\lambda_{HS_1}^{(A)}\left(2T_\nu^{(1)}+6\,\mathrm{Tr}\Big(Y_d\,\left(Y_d\right)^\dagger\Big)+2\,\mathrm{Tr}\Big(Y_e\,\left(Y_e\right)^\dagger\Big)+6\,\mathrm{Tr}\Big(Y_u\,\left(Y_u\right)^\dagger\Big)\right)$ | $+\frac{8}{3}\,\mathrm{Tr}\Big(y_1\,\left(y_1\right)^\dagger\,Y_e\,\left(Y_e\right)^\dagger\Big)$ | $+\frac{16}{3}\lambda_5\left(\lambda_5\right)^*$ |

Quantity:  $\lambda_{HS_1}^{(A)}$

Terms 2 of 3

| Column | Representative term structure                     |
|--------|---------------------------------------------------|
| T1     | $+g_1^2\left(6g_2^2-3\lambda_{HS_1}^{(A)}\right)$ |
| T2     | $-9g_2^2\lambda_{HS_1}^{(A)}$                     |
| T3     | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                | T2                            | T3                                        |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|---------------------------------------------------|-------------------------------|-------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+g_1^2\left(6g_2^2-3\lambda_{HS_1}^{(A)}\right)$ | $-9g_2^2\lambda_{HS_1}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |

Quantity:  $\lambda_{HS_1}^{(A)}$

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $+2\lambda_1\lambda_{HS_1}^{(A)}$            |
| T2     | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ |
| T3     | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                | T2                                           | T3                                          |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------------------|----------------------------------------------|---------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |

$$\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

**Quantity:**  $\lambda_{HS_2}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(A)}}{d\mu}=\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                                                                                                                                              |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{HS_2}^{(A)}\left(2T_\nu^{(2)}+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\right)$ |
| T2     | $+8\,\text{Tr}\Big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$                                                                                                            |
| T3     | $+16\lambda_5\,(\lambda_5)^*$                                                                                                                                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                                                                                                                                         | T2                                                              | T3                            |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $\lambda_{HS_2}^{(A)}\left(2T_\nu^{(2)}+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\right)$ | $+8\,\text{Tr}\Big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$ | $+16\lambda_5\,(\lambda_5)^*$ |



Quantity:  $\lambda_{HS_2}^{(A)}$

Terms 2 of 3

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $+g_1^2\left(18g_2^2-15\lambda_{HS_2}^{(A)}\right)$ |
| T2     | $-9g_2^2\lambda_{HS_2}^{(A)}$                       |
| T3     | $+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                                  | T2                            | T3                                        |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------------------------------------|-------------------------------|-------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+g_1^2\left(18g_2^2-15\lambda_{HS_2}^{(A)}\right)$ | $-9g_2^2\lambda_{HS_2}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$ |

Quantity:  $\lambda_{HS_2}^{(A)}$

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $+2\lambda_1\lambda_{HS_2}^{(A)}$            |
| T2     | $+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$ |
| T3     | $+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$ | T1                                | T2                                           | T3                                          |
|-------|----------|-----------|---------------|-----------|---------------|-------|-------|-----------------------------------|----------------------------------------------|---------------------------------------------|
| T3-B  | 1        | 2         | $\frac{1}{2}$ | 2         | $\frac{3}{2}$ | 1     | 1     | $+2\lambda_1\lambda_{HS_2}^{(A)}$ | $+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$ | $+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$ |